



Short time cardio-vascular pulses estimation for dengue fever screening via continuous-wave Doppler radar using empirical mode decomposition and continuous wavelet transform

Nguyen Dinh Chinh^b, Luu Manh Ha^{a,*}, Guanghao Sun^c, Le Quoc Anh^a, Pham Viet Huong^b, Tran Anh Vu^d, Tran Trong Hieu^a, Tran Duc Tan^e, Nguyen Vu Trung^{f,g}, Koichiro Ishibashi^c, Nguyen Linh Trung^a

^a University of Engineering and Technology, Vietnam National University, Hanoi (UET-VNU), Viet Nam

^b International School, Vietnam National University, Hanoi (IS-VNU), Viet Nam

^c The University of Electro-Communications (UEC), Japan

^d Hanoi University of Science and Technology (HUST), Viet Nam

^e Faculty of Electrical and Electronic Engineering, Phenikaa University, Hanoi 12116, Viet Nam

^f National Hospital of Tropical Diseases (NHTD), Viet Nam

^g Hanoi Medical University (HMU), Viet Nam

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ABSTRACT

Contactless measurement of cardio-vascular pulse acts an essential role in clinical medical sectors. Estimating cardio-vascular pulses based on continuous-wave (CW) Doppler radar in limited time while maintaining the accuracy is a challenging task. In this paper, we propose a signal processing method that combines empirical mode decomposition (EMD) and continuous wavelet transform (CWT) for a short time estimation of heart rate (HR) and inter-beat-interval from radar signals. We evaluate performance of the proposed method using 85 patients with dengue fever and 40 healthy subjects. Subsequently, the estimated contactless HR is compared to that of a commercial contact-type medical device. The result shows that the HR can be estimated within a period of 5 s with an accuracy of $96.2 \pm 2.5\%$. The patients with dengue fever show an elevated HR and a decreased standard deviation of heartbeat interval (SDHI). Finally, linear discriminant analysis (LDA) is utilized with two parameters HR and SDHI to classify the dengue patients, achieving the sensitivity and specificity values of 86.3% and 86.9%, respectively.

1. Introduction

Dengue is an infectious disease, commonly occurs in tropical and subtropical countries, which has been a global threat in the first two decades of the 21st century [1]. It is estimated that there are 390 million cases of the annual occurrence [2]. Hence, dengue screening is important to minimize the possibility of dengue fever outbreaks. Body temperature measurement devices are widely used in public areas for the screening; however, several recent studies have indicated that changing body temperature is not only caused by fever symptoms but also by antipyretic agents and ambient temperature, and thus it is insufficiently reliable [3,4]. Beside body temperature, heart rate is another important information for the infection screening. There have been several studies

reported of rapid HRs while having the symptoms of dengue fever and of decreasing HRs during recovery [5,6]. Moreover, some recent studies have suggested that there is a difference in standard deviation of heartbeat interval (SDHI) of dengue patient and healthy person [7,8].

Although contact method for HR measurement such as electrocardiography (ECG) and photoplethysmography (PPG) have shown highly reliable results [9], they may contain several drawbacks in clinical practice. For instance, the contact methods may be limited due to privacy and inconvenient use, making them difficult to be applied at a large scale. In addition, the clinical staffs have to reach the patients for the device attachment, potentially causing them being exposed to a high risk, especially in case of highly contagious infections. Therefore, research for contactless measurement methods for HR detection has

* Corresponding author.

E-mail address: halm@vnu.edu.vn (L.M. Ha).

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been highly active, such as in public healthcare. In screening infectious diseases such as dengue fever, a contactless system, which is based on Doppler radar, can be used to detect HR and to classify infected patients [10]. The principle of Doppler radar-based screening is that the wave is capable to penetrate through objects such as clothes, and then rebounds with the information of the displacement of wall chest induced by the heart activity and the respiratory motion [11,12]. The contactless system, based on CW Doppler radar, has several advantages such as low power consumption, large detection range and simple radio architecture [11,13]. Several researches on HR estimation using Doppler radar have been published recently. Ohtsuki et al. [14] proposed a method based on estimating the average R-R interval applying the Viterbi algorithm for heartbeat detection; yet, the study assumed that the R-R intervals are almost equal within a short period, which is not appropriate for the cases with varying R-R interval. Yu et al. [15] introduced a method using higher order cyclostationary (HOCS) and proved that the third-order cyclic cumulant can detect the HR. Le [16] proposed a method to improve the SNR of the heartbeat signal based on high order cumulant (HOC). Subsequently, Z-chirp transform was applied to zoom in a specific band of frequency to extract the HR. Saluja et al. [17] applied a gamma filter, a supervised machine learning algorithm, to eliminate the respiratory signal and then extract the heartbeat signal; nevertheless, it is just feasible to extract the HR from the radar signals of arbitrary subjects with reasonable accuracy when a vast amount of data are involved in training the model.

HR estimation in a short time is crucial in several clinical applications, including estimating the vital signs of patient at the registration desk in health care facilities (especially worthy in emergency cases), and fast screening passengers from epidemic areas which may indicate the probability of infection, requiring a short determination time (typically within 5 s or less) [18]. Some conventional methods such as Fourier transform (FFT), HOCS and HOC are not suitable for fast estimation of cardio-vascular pulse, because these methods require a period of signal longer than 10 s to convert to the frequency domain and highlight the frequency composition of the heartbeat. Some recent studies have been focused on cardio-vascular pulse estimation in a short time using Doppler radar signal. Yang et al. [19] utilized a peak-detection algorithm in the time domain to measure the HR extracted from the radar signal by applying two band pass filters (BPF) in approximately 5 s, and obtained the correlation coefficient of 0.92 between the raw extracted heartbeat signal and a reference ECG. Furthermore, several methods based on data-length-variation techniques have been proposed to estimate the HR in a short time. Tu and Lin [20] studied 4 healthy subjects using FFT and showed that the HR can be determined within time variation of from 2 to 5 s with an error of 3.4%. Li and Lin [18] applied a method with the wavelet transform coefficients on data from 3 subjects, and achieved an error of less than 4% within window times from 3 to 5 s. Those mentioned methods, however, analyze the HR in the frequency domain and thus are not suitable to extract SDHI information. In addition, the mentioned studies just carried on a limited amount of healthy subjects, and thus their results are still unfirmly solid with dengue patients.

Typically, the displacement of the wall chest caused by the breathing is naturally often stronger than the displacement of the heart activity [18,21]. Also, CW Doppler radar is widely confirmed that the received signal has non-stationary and non-linear characteristics [22]. As the result, separating the heartbeat signal from the output of continuous-wave Doppler radar is challenging and garners the attention of researchers. Based on the fact that heartbeat signal can be found in the specific frequency range from 0.5 to 2.5 Hz [23], there have been several studies related to heartbeat signal separation using BPF [19,24], wavelet transform (WT) [18,25] and empirical mode decomposition (EMD) [26]. Droitcour et al. [24] used a two-stage BPF, in which, the first stage BPF is to reduce high-frequency noise while the second stage is to extract the heartbeat signal. While BPF is not suitable method for processing the non-linear and non-stationary signal, WT and EMD may be used to

handle this problem. He and Liu [25] proposed a method utilizing WT with the frequency range from 0.8 to 1.5 Hz, and the heartbeat signal retrieved by wavelet reconstruction. Mostafanezhad et al. [26] applied an EMD-based method to eliminate high frequency motion noise. Specifically, the received signal has a large respiratory amplitude and motion noise, thus a high-pass filter was used to attenuate the respiratory frequency before utilizing the EMD method for the heartbeat signal extraction. Subsequently, a window-based FFT method was applied to estimate the HR. Besides, Hu and Jin [27] used an improved ensemble empirical mode decomposition (EEMD) to remove the high frequency noise in the signal and then applied CWT to extract the heartbeat presented by a single frequency scale. Principle component analysis (PCA) has been utilized to extract cardio-vascular pulse patterns from the received signal [28]. PCA, however, is not suitable for signals that contain non-Gaussian noises, especially the burst noise. To the best of our knowledge, considering the paucity of studies on short time cardio-vascular pulses estimation on dengue fever infection screening, it is necessary to investigate insightfully on this problem.

In this study, we aim to classify dengue patients in a short period using the HR and SDHI information extracted by processing signal from CW Doppler radar. Constitutively, we propose a coarse-to-fine approach that combines the EMD and CWT to estimate the cardio-vascular pulse via CW Doppler radar during a short period. EMD is used for decomposing natural signals into intrinsic mode functions (IMFs). Hence, it is able to decompose and estimate components of short sample length accurately in the time domain, which can separate the heartbeat from the signal [26,29]. In addition, due to the fact that the heartbeat signal can be distorted, CWT is used to provide a time-frequency map of the signal being analyzed with high frequency resolution and sharp time resolutions [30]. In this work, EMD is firstly used to decompose the I/Q signals into IMFs. Next, the most resemble IMF to the reference heart beat signal is selected based on the maximum of the normalized cross-correlation between each IMF and the reference signal in a pilot study. Afterwards, in order to eliminate the incorrect heart peaks in the chosen IMF, a scale-limited CWT reconstruction is applied to reform a new signal (reconstructed IMF). Subsequently, based on peak detection algorithm [31] on the reconstructed signal, the HR and SDHI information can be extracted in the time domain. Finally, we apply linear discriminant analysis (LDA) method to classify dengue patients versus healthy subjects using 85 clinical data from hospital and 40 healthy volunteers.

The rest of this paper is organized as follows. In Section 2 the material and method section, we introduce the contactless system used for data acquisition and the proposed signal processing method to estimate cardio-vascular pulses and the HR from the received signals. Section 3 presents evaluation results of the proposed method, compared to HR extraction by a conventional method and from a commercial contact device. Also, classification of the dengue patients using LDA with the HR and SDHI is mentioned in this section. These achieved results are discussed in Section 4 while conclusions are drawn in Section 5.

2. Material and method

2.1. CW Doppler radar

The theory of contactless measurement of cardio-vascular pulses based on CW Doppler radar was described in several publications [12, 18–20]. Fig. 1 shows the operating mechanism for measuring the vital signs using CW Doppler radar.

Generally, the Doppler radar device transmits the output continuous wave $T(t)$ to the wall chest:

$$T(t) = A_T \cos[2\pi f t + \phi(t)], \quad (1)$$

where A_T is the amplitude of transmitting signal, f is the frequency of the transmitting signal, and $\phi(t)$ is the phase noise.

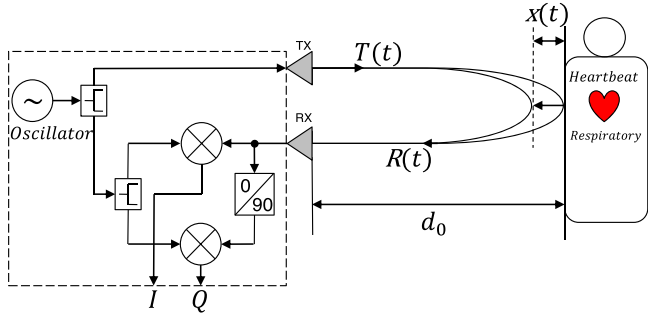


Fig. 1. Operating mechanism of the contactless system using CW Doppler radar with I/Q signal for cardio-vascular pulses detection.

By the Doppler principle, the reflected signal from the human body is modulated by the movement of the body wall chest with the displacement of $x(t)$ driven by the heartbeat and respiratory circle. At the receiver, the reflected signal $R(t)$ is given by:

$$R(t) = A_R \cos \left[2\pi f t - \frac{4\pi d_0}{\lambda} - \frac{4\pi x(t)}{\lambda} + \phi \left(t - \frac{2d_0}{c} \right) \right], \quad (2)$$

where A_R is the amplitude of the received signal, λ is the wavelength, c is the speed of light, and d_0 is the distance between the Doppler radar and the body surface. After $R(t)$ is down-converted, two baseband signals are obtained, which are the in-phase signal $I(t)$ and the quadrature phase signal $Q(t)$ as below:

$$I(t) = A_I \cos \left[\frac{4\pi d_0}{\lambda} - \frac{4\pi x(t)}{\lambda} + \phi(t) + \phi \left(t - \frac{2d_0}{c} \right) \right], \quad (3)$$

$$Q(t) = A_Q \cos \left[\frac{4\pi d_0}{\lambda} - \frac{4\pi x(t)}{\lambda} + \phi(t) + \phi \left(t - \frac{2d_0}{c} \right) + \frac{\pi}{2} \right], \quad (4)$$

where A_I and A_Q are the amplitude of the in-phase signal and the amplitude of quadrature phase signal.

In our experiment, a CW Doppler radar (New-JRC, Tokyo, Japan) was used to produce the baseband signals $I(t)$ and $Q(t)$. The device operates using 10 GHz radar wave with effective isotropic radiated power of 20 mW. The output signals were passed through a 12-bit analog-to-digital converter (National Instruments, USB-6008) with a sampling frequency of 100 Hz and a bandwidth of 0.1–50 Hz. The measurement flow was controlled using LABVIEW. In this study, we used a PPG pulse oximeter (SAT-2200, NIHON KOHDEN, Tokyo, Japan) for the reference signal acquisition. The cardio-vascular pulse signal was processed to extract the reference signal to compute the reference HR.

2.2. Data collection

In laboratory experiments, 40 data of ordinary people (32 males and 8 females, 18–40 years old) identified without cardiac diseases or other related diseases recently involved in the data acquisition. The measurement time for each subject maintained at 30–120 s. All ordinary subjects were students of University of Electro-Communication (UEC) and University of Engineering and Technology (UET) of Vietnam National University, Hanoi.

We performed the experiment on the clinical data which were acquired in our previous study [10]. The data from a total of 410 patients (225 males and 185 females, 15–77 years old) were collected at the National Hospital of Tropical Disease (NHTD), Hanoi, Vietnam in 2017 using the contactless system. Of those, 369 were diagnosed with dengue by a rapid serological test. The acquiring time for each subject is 30–60 s. The Doppler radar was placed at position with approximate distance of 0.2 m from the subject. This study was approved by the Ethics Committee of the NHTD and UEC. All subjects accepted their data for research purposes with an informed consent.

In preprocessing step, 12 pilot data (6 dengue patients and 6 healthy subjects) were randomly selected for optimizing the parameters of the proposed method. Subsequently, we selected 306 visually stable segments the signals from the dataset of the two groups (85 dengue patients and 40 healthy subjects), 153 segments for each, for qualitatively evaluating the method and testing the dengue classification.

2.3. Proposed signal processing method

The diagram of the proposed method for short time cardio-vascular pulses estimation is illustrated in Fig. 2. Firstly, the CW Doppler radar output signal, either $I(t)$ or $Q(t)$, was segmented by a moving Hamming window of 5-s data length with a step of 0.5 s, resulting in $I_{5s}(t)$ or $Q_{5s}(t)$ signal correspondingly. Secondly, the windowed signal was decomposed by using EMD method to obtain the IMF components. Subsequently, CWT was applied to extract components belonging to the frequencies of the heartbeat. Finally, the peaks of the cardio-vascular pulses were detected in the time domain, and then the interval from the first peak to the last peaks was used to calculate the HR.

2.3.1. Empirical mode decomposition

Each segment of the CW Doppler radar output signal is decomposed by using EMD as in the coarse step. EMD [32] is a fully data driven technique used widely to decompose both non-stationary and nonlinear signal into narrow band elements named IMFs, denoted as $c_i(t)$ in the following equation:

$$X(t) = \sum_{i=1}^n c_i(t) + r_n(t), \quad (5)$$

where $X(t)$ is the signal which is decomposed and $r_n(t)$ is the last residue, which can be either the mean trend or a constant, and n is the number of IMFs $c_i(t)$. IMFs $c_i(t)$ are determined from $X(t)$ in a sifting process [26].

The decomposing process stops when either the maximum desirable number of IMFs is reached or Cauchy-type stop criterion D of residue $r_n(t)$ is satisfied as the following [33]:

$$D \triangleq \left\| \frac{r_{\text{prev}}(t) - r_{\text{cur}}(t)}{r_{\text{prev}}(t)} \right\|_2 < \varepsilon, \quad (6)$$

where $r_{\text{prev}}(t)$ is the previous residue and $r_{\text{cur}}(t)$ is the current residue, typically ε is a constant of 0.2 or 0.3. In this study, the stop condition of EMD is the maximum number of IMFs extracted. Based on an experiment of the pilot data, we achieved the maximum number of IMFs of 4 (see Fig. 3).

As mentioned in Section 2.3, in order to remove noise in the raw reference signal, i. e. the PPG signal, we applied a 3rd-order low-pass Butterworth filter with a cut-off frequency of 2.5 Hz. The filtered signal was later used as reference signal in this study (see Fig. 3j).

To verify which IMF is suitable to estimate the cardio-vascular pulses, we computed the cross-correlation between the reference signal and each IMF component. Given an IMF $x(t)$ and a reference heartbeat signal $y(t)$, the cross-correlation of between $x(t)$ and $y(t)$ is calculated as the following formula:

$$R(\tau) = x(t) * y(t + \tau) = \int_{-\infty}^{+\infty} \overline{x(t)} y(t + \tau) dt, \quad (7)$$

where $*$ denotes correlation, $\overline{x(t)}$ represents the complex conjugation, and τ is the time lag. The cross-correlation coefficient is normalized by the following:

$$\widehat{R}(\tau) = \frac{R(\tau)}{C(0)}, \quad (8)$$

where $\widehat{R}(\tau)$ is the normalized cross-correlation, $C(0)$ is the autocorrelation of the reference signal at zero lag.

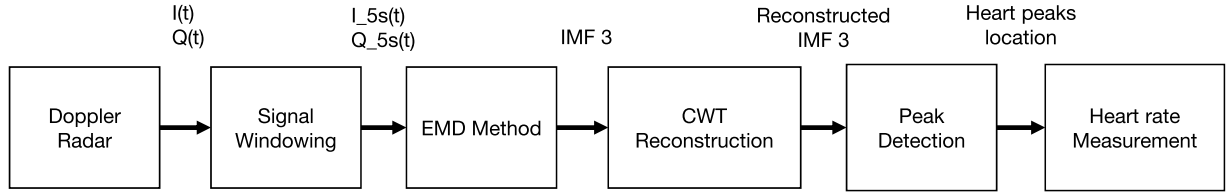


Fig. 2. Block diagram of the proposed method for short time HR estimation via the CW Doppler radar system using combination of EMD and CWT.

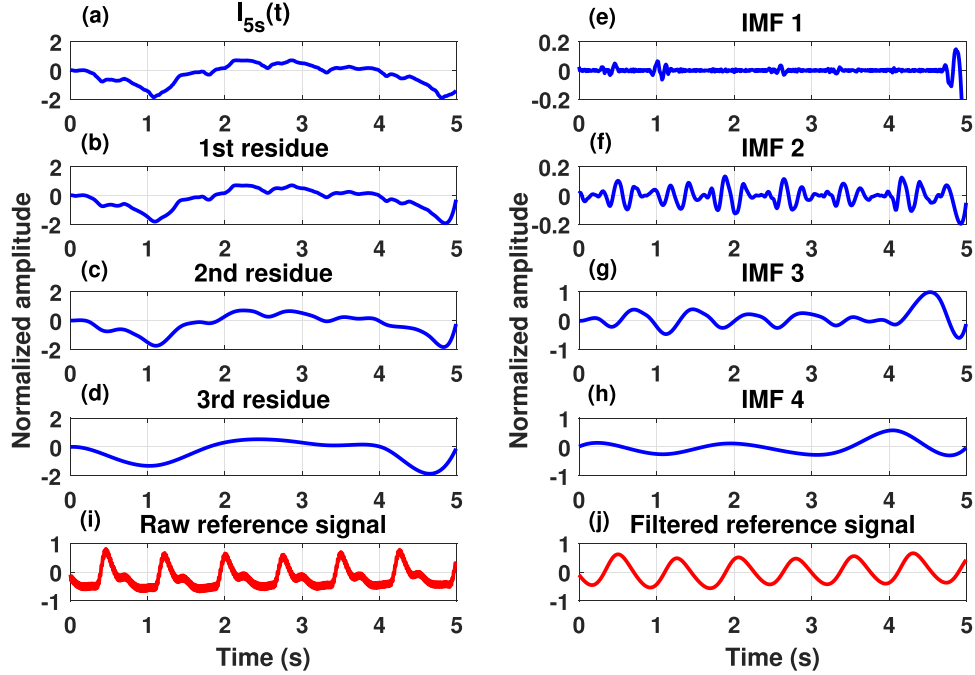


Fig. 3. An example of a 5-s data length analysis: (a) raw data from contactless system; (b–d) residue components after IMFs extracted; (e–h) IMFs extracted after EMD method; (i) raw reference signal; (j) reference signal after filtered.

In the experiments on the pilot data, the mean value of the maximum normalized cross-correlation (MNCC) coefficient of 0.78 ± 0.19 between IMF3 and the filtered reference signal (see Table 1) indicates that IMF3 is most similar to the reference signal. Therefore, it can be confirmed that IMF3 is the most suitable candidate to estimate the heartbeat. In later experiments on the test dataset, we use the radar signals as standalone information for estimating the HR and only use the reference signal for evaluating the accuracy of the HR estimation.

2.3.2. Continuous wavelet transform

The continuous wavelet transform analyzes a signal by converting it from the time domain into the time-frequency domain [18]. Its superiority is its ability to moderate the scale and position parameters of the wavelet function, which enables wavelet transform to be suitable for signal analysis in short time. In this study, a reconstruction wavelet transform was applied to IMF3. Because IMF3 may be distorted (see Fig. 4), the distortion period contains frequency components outside the

range of the heartbeat. The scale-limited wavelet reconstruction, thus, is to correct the signal form by removing the redundant wavelet components. The Morlet wavelet, defined as $e^{-x^2/2} \cos 5x$, was used as an analyzing function. We chose it as the analyzing function for two reasons. Firstly, it contains a single frequency component while a cardio-vascular pulse is periodic, and generally can be found at the specific frequency range of 0.5–2.5 Hz, and thus the Morlet wavelet function can be rescaled to match with the heartbeat frequency, ensuring an accurate record of the heartbeat signal. Secondly, the shape of the Morlet wavelet function resembles the IMF3 component, thus it can be matched to the component by moderating the scale and position parameters.

The formula of wavelet transform of a signal $f(t)$ is shown as follows:

$$W_{\psi}f(a, b) = \int_{-\infty}^{+\infty} f(t) \overline{\psi_{a,b}} dt, \quad (9)$$

where a is the scale parameter, b is the position parameter, and ψ is the analyzing wavelet function.

The scale parameter a enables a useful property of the analyzing wavelet function, which relates to the frequency. The reconstruction formula takes place for all $f \in L_2(\mathbb{R})$:

$$f(t) = \frac{1}{C_{\psi}} \int_a \int_b \psi_{a,b}(t) W_{\psi}f(a, b) \frac{da db}{a^2}, \quad (10)$$

where C_{ψ} is an admissible constant [34].

In this task, the wavelet reconstruction in the range of frequency of

Table 1

MNCC coefficients between the IMFs and the filtered reference signal.

IMFs	MNCC coefficients
IMF1	0.08 ± 0.06
IMF2	0.25 ± 0.15
IMF3	0.78 ± 0.19
IMF4	0.32 ± 0.18

The bold values significance the largest value.

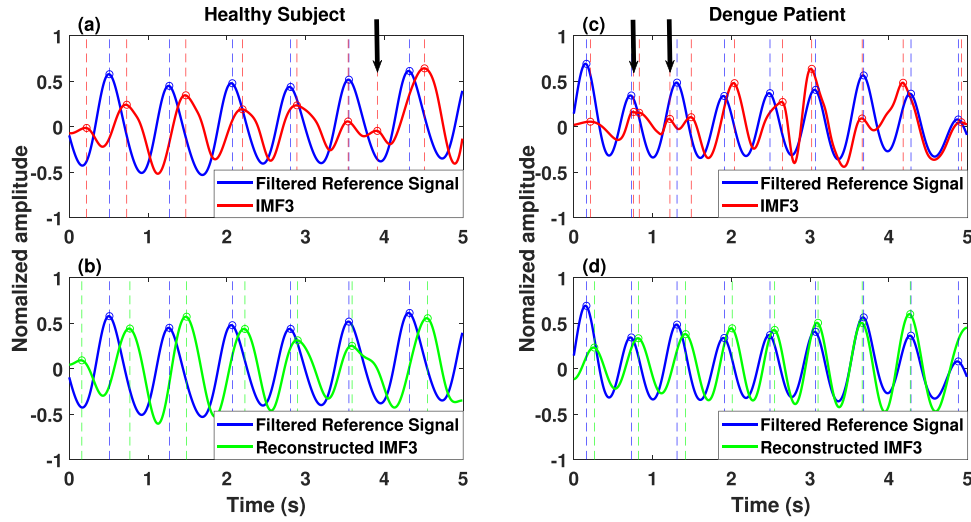


Fig. 4. (a, c) The peak locations of the IMF3 component and the reference signal while (b, d) the wavelet reconstructed IMF3. The arrows point at position of the incorrect heart peaks in the IMF3 components. The left figures are the components from of a healthy subject and the right one are the components from a dengue patient.

between 0.5 and 2.5 Hz, which is the range of the heartbeat frequencies [23], is applied on IMF3. Subsequently, the heart peaks from the reconstructed IMF3 can be extracted by using the peak detection algorithms [31]. We performed an analysis on the wavelet reconstructed IMF3 on the pilot data by computing the normalized cross-correlation between the reconstructed signal and the reference signal, achieving the mean of the MNCC coefficient of 0.85 ± 0.12 (higher than that of the original IMF3).

Fig. 4 shows two examples of wavelet reconstruction of IMF3 estimate from the healthy subject and the dengue patient. It can be seen that at the arrow positions, the IMF3 contains incorrect heart peaks meanwhile those are eliminated in the wavelet reconstructed IMF3.

Subsequently, the HR can be estimated by using the following formula:

$$HR = \frac{60 \cdot (n_{HR} - 1)}{l_{last} - l_{first}}, \quad (11)$$

where n_{HR} is the number of detected peaks; l_{last} and l_{first} are the time location of the last peak and the first peak, respectively.

Based on the detected heart peak positions, the heartbeat intervals can be determined (see in Fig. 5), then we compute the SDHI by the following formula:

$$s^2 = \frac{1}{n_{HI} - 1} \sum_{i=1}^{n_{HI}} (x_i - \bar{x})^2, \quad (12)$$

where s is the SDHI, n_{HI} is number of heartbeat interval, $\bar{x}$ is the average of the heartbeat interval and x_i is the “ith” heartbeat interval.

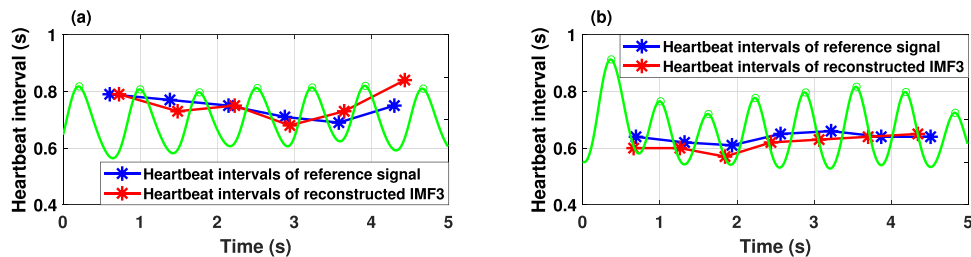


Fig. 5. Examples of the heartbeat intervals of a healthy subject (a) and a dengue patient (b) with the corresponding reference signals (the green lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3. Experiments and results

3.1. Heart rate measurement

In this section, we performed the HR estimation using the proposed method, and computed the accuracy w.r.t the periods from 3 to 5 s on the pilot dataset. The accuracy of the estimated HR was define as:

$$\text{Accuracy} = \frac{|HR_{est} - HR_{ref}|}{HR_{ref}} \cdot 100\%, \quad (13)$$

where HR_{est} is the estimated HR and HR_{ref} is the reference HR.

The result showed that the period of 5 s yields the smallest error value, compared to that of 4 and 3 s, which has the error value of $4.3 \pm 3.7\%$, $10.6 \pm 9.8\%$ and $12.7 \pm 11.9\%$, respectively (see Table 2). Based on this result, we used 5-s data-length for the short time HR estimation in later experiments.

We compared the estimated HR by using the EMD-CWT method, that by using the conventional method, short time Fourier transform (STFT) [35], and the reference HR obtained from the reference signal by using Eq. (10). EMD-CWT method achieved higher accuracy as compared to STFT, yielding the average of error rate, within the 95% outlier limit, of $4.3 \pm 3.7\%$ and $26.9 \pm 25.3\%$, respectively. Fig. 6 illustrates examples of the estimated HR from a healthy and a dengue subject respectively, using EMD-CWT, STFT and the reference HR at each time point of 5 s.

Next, accuracy of the HR estimation by the proposed method, evaluated on all of 125 data with 306 selected stable segments, is illustrated in Fig. 7. Fig. 7a shows the scatter chart of the correlation coefficient between the estimated HR and the reference HR; the correlation coefficient is $R = 0.972$. We calculated the relative accuracy of HR

Table 2

Percentage errors and absolute errors of the estimated HR on the pilot dataset w.r.t periods of time.

Data number	Percentage errors			Absolute errors (bpm)		
	3 s	4 s	5 s	3 s	4 s	5 s
1 (healthy)	$8.4 \pm 6.7\%$	$6.2 \pm 7.1\%$	$3.4 \pm 2.5\%$	5.6 ± 4.3	4.1 ± 4.6	1.7 ± 1.6
2 (healthy)	$9.5 \pm 8.7\%$	$7.3 \pm 8.9\%$	$2.8 \pm 2.4\%$	7.7 ± 7.1	5.9 ± 7.1	2.3 ± 2.0
3 (healthy)	$11.9 \pm 10.4\%$	$12.1 \pm 11.2\%$	$4.2 \pm 2.9\%$	9.5 ± 8.4	9.7 ± 9.1	3.4 ± 2.3
4 (healthy)	$8.4 \pm 7.6\%$	$7.4 \pm 8.2\%$	$3.1 \pm 2.5\%$	6.1 ± 5.5	5.3 ± 5.9	2.3 ± 1.9
5 (healthy)	$11.7 \pm 10.4\%$	$10.4 \pm 9.4\%$	$3.1 \pm 2.7\%$	7.8 ± 6.9	6.9 ± 6.1	1.9 ± 2.1
6 (healthy)	$13.0 \pm 9.9\%$	$14.5 \pm 10.8\%$	$4.1 \pm 3.6\%$	10.6 ± 8.1	11.8 ± 8.8	3.7 ± 3.2
7 (dengue)	$14.5 \pm 9.4\%$	$13.1 \pm 8.8\%$	$6.5 \pm 3.4\%$	15.7 ± 10.3	14.1 ± 9.4	6.9 ± 3.6
8 (dengue)	$15.4 \pm 11.4\%$	$13.7 \pm 8.2\%$	$6.7 \pm 4.6\%$	15.7 ± 11.6	13.9 ± 8.4	6.8 ± 4.6
9 (dengue)	$15.2 \pm 11.9\%$	$9.9 \pm 5.3\%$	$3.0 \pm 2.4\%$	15.1 ± 12.0	9.8 ± 5.3	3.4 ± 2.5
10 (dengue)	$13.9 \pm 9.7\%$	$13.8 \pm 8.9\%$	$6.4 \pm 5.9\%$	17.5 ± 12.2	16.9 ± 11.1	8.0 ± 7.4
11 (dengue)	$5.6 \pm 6.4\%$	$4.5 \pm 3.5\%$	$3.3 \pm 3.1\%$	6.1 ± 7.1	4.9 ± 3.9	3.6 ± 4.7
12 (dengue)	$25.1 \pm 21.5\%$	$16.3 \pm 15.3\%$	$6.6 \pm 5.3\%$	22.5 ± 18.9	14.7 ± 13.4	6.2 ± 5.6
Total	$12.7 \pm 11.9\%$	$10.6 \pm 9.8\%$	$4.3 \pm 3.7\%$	11.7 ± 11.3	9.7 ± 9.0	4.1 ± 3.8

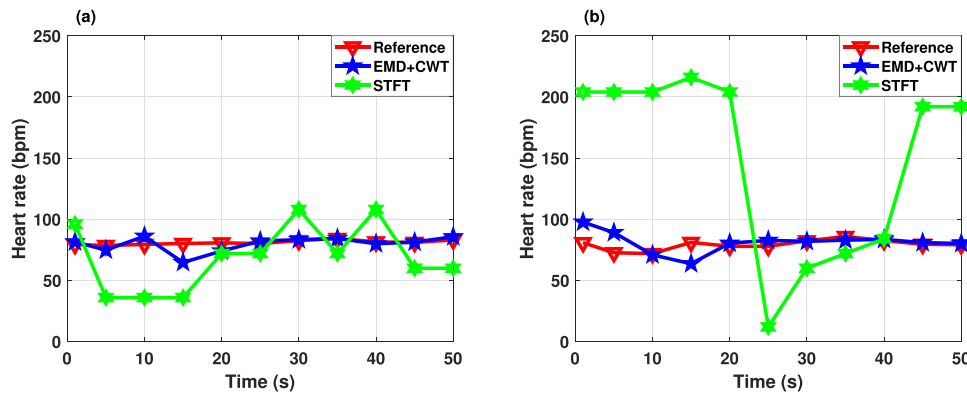


Fig. 6. Example of the HR estimated by the proposed method and STFT in an interval of 5 s w.r.t time points. The red line is the reference HR; the blue line is the HR estimated by the proposed method, and the green line is the HR by STFT. (a) A case of the healthy subjects, and (b) a case of the dengue patients. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

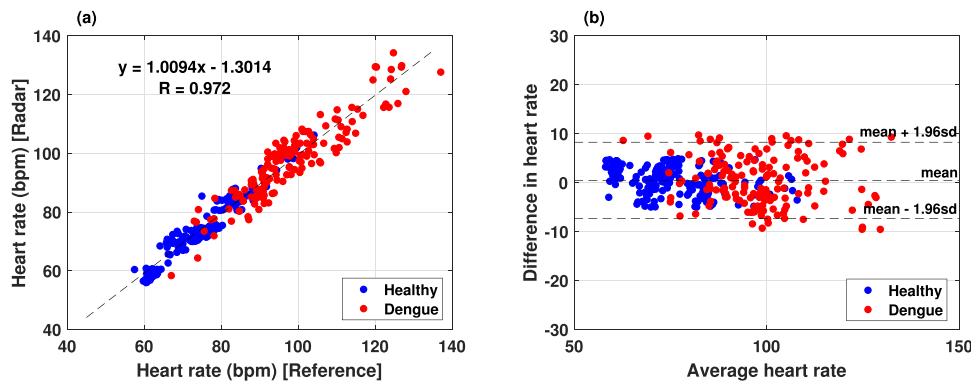


Fig. 7. Estimated heart rate by the proposed method versus the reference heart rate; (a) scatter chart; (b) Bland-Altman chart.

estimation versus the reference HR, and achieved an accuracy of $96.2 \pm 2.5\%$. Moreover, the errors in HR estimation on the healthy subject's dataset and the dengue patient's dataset are of $3.3 \pm 2.1\%$ and $4.4 \pm 2.8\%$, respectively. Fig. 7b presents the Bland-Altman chart of difference between the estimated HR and the reference HR with the 95% limits of agreement range from -7.28 to 8.28 bpm (mean $\mu = 0.5$). It can be seen in Fig. 7b that the difference of the dengue patients has a larger standard deviation ($\sigma = 5.08$ bpm) than that of the healthy subjects ($\sigma = 2.76$ bpm).

3.2. Data analysis for dengue patient classification

The distributions of the HR of the healthy subjects and the dengue patients, computed by the proposed method, are illustrated in Fig. 8. We found that the average of the HR of the dengue patients is higher than that of the healthy subjects. Specifically, the average of the HR of the dengue patients is 95 beats per minute (bpm), while that of the healthy subjects is 80 bpm.

Fig. 8b shows the ROC curve of using only the HR information to identify the dengue patients. The sensitivity and specificity of the classification are calculated using the following formulas:

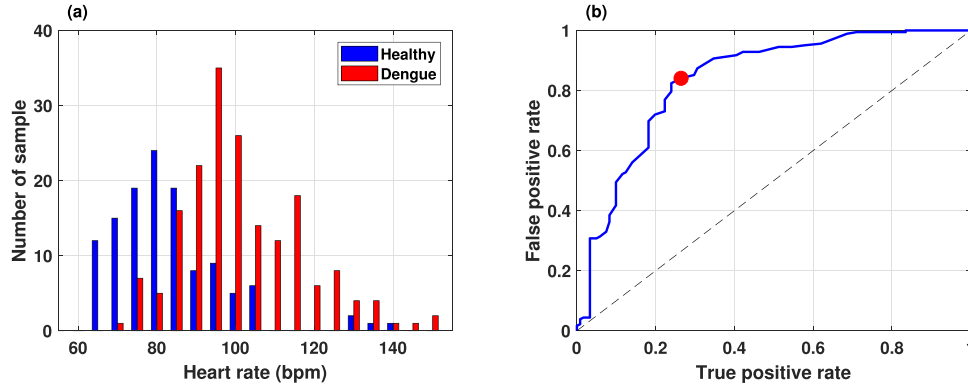


Fig. 8. Statistical analysis using HR distribution of the both dengue patients and healthy subjects (a), and ROC curve of the classification (b).

$$\text{Sensitivity} = \frac{TP}{TP + FN}, \quad (14)$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \quad (15)$$

where TP is the number of correctly identified the dengue patients, TN is the number of correctly identified healthy subjects, FN is the number of incorrectly identified healthy subjects while FP is the number of incorrectly identified dengue patients. The sensitivity is the proportion of the dengue patients who are classified as positive while the specificity indicates the proportion of the healthy subjects who are classified as negative. At the optimal threshold of 87 bpm (the dot in Fig. 8b), the sensitivity and specificity of the classification are 82.4% and 76%, respectively.

Furthermore, we performed an analysis for dengue patients classification using both of the HR and SDHI information. Fig. 9 presents the boxplots of the HR and SDHI of the dengue patients and healthy subjects extracted from the radar signal and the reference signal. It can be seen that in general, the SDHI of the dengue patients is smaller than that of the healthy subjects (Fig. 9b). This result is similar to that stated in the study by Yang et al. [36].

Based on the hypothesis that both of the HR and SDHI provides independent information for classifying dengue patients, we applied t -test on the HR and SDHI between the dengue patients and healthy subjects. The result showed that both p -values are smaller than 0.001, meaning that the HR and SDHI between the dengue patients and healthy subjects are statistically significantly different. Specifically, the medians of HR of the dengue patients and healthy subjects were 99.1 and 72.7 bpm, respectively. Meanwhile, the medians of the SDHI of the dengue patients and healthy subjects were 17.6 and 40.3 ms, respectively.

In this experiment, we used LDA [37], a common method used in biomedical data processing [38], to classify the dengue patients and

healthy subjects using two parameters: HR and SDHI. Specifically, LDA is used to find a linear projection so that the proposition between the between-class variance and the within-class variances of the two groups is maximized. As a result, the discriminate function f for the classification is derived as the following:

$$f(\text{HR}, \text{SDHI}) = -0.323 + 0.0131 \cdot \text{HR} - 0.0096 \cdot \text{SDHI}, \quad (16)$$

From the discriminate function performed on the whole dataset, we obtained ROC curve of the classification as shown in Fig. 10a. Subsequently, the optimal threshold of $f = 0.55$ was determined from the ROC curve, yielding the sensitivity and the specificity of 86.3% and 86.9% respectively. Fig. 10b presents scatter graph of HR and the SDHI of the both the dengue patients and healthy subjects. The classification line (the green line), derived from determining the optimal discriminate function value, indicates the border of the dengue patient and healthy subject clusters.

4. Discussion

In this study, we proposed a signal processing method, which combines EMD and CWT, to solve the problem of the short time HR estimation from the non-linear, non-stationary CW Doppler radar signal. The experiment with several periods of time indicated that the proposed method is able to estimate the HR within 5 s. Afterwards, the accuracy of the proposed method was compared to that of a conventional method STFT. In addition, the evaluation was performed on the data from 85 dengue patients and 40 healthy subjects. Finally, the HR and SDHI information were used to classify the dengue patients using LDA method.

In Table 2, we can see that reducing the window length for the HR measurement results in increasing in the error rate of the HR estimation. The reason could be that when the window length is small, the wavelet transform does not have sufficient wavelet coefficients to reconstruct the heartbeat signal. We also found that the window length of 5 s provides

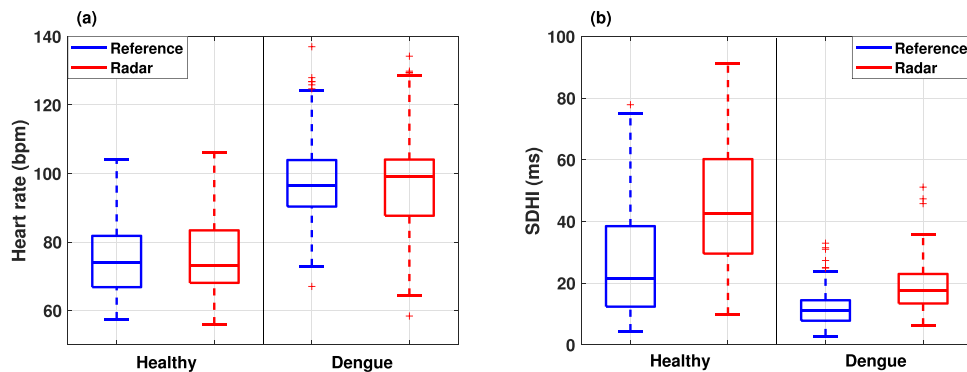


Fig. 9. Boxplots of the HRs (a) and the SDHIs (b) of the healthy subjects and dengue patients from the CW Doppler radar signal and the reference signal.

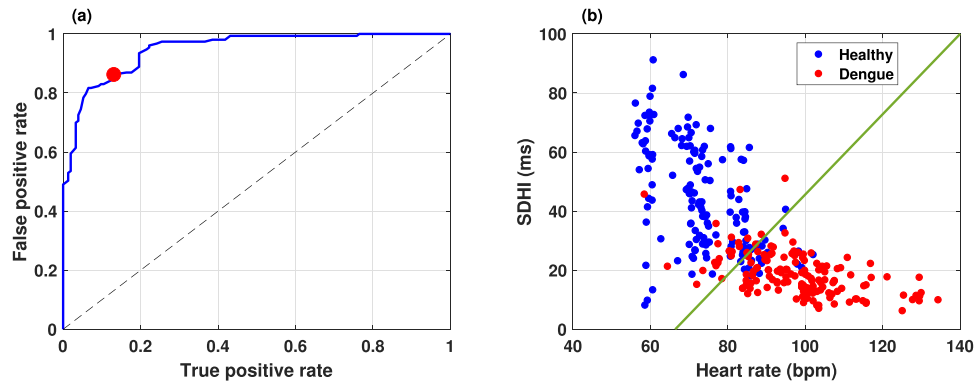


Fig. 10. Classification of the dengue patients using LDA method (a) ROC curve for the classification with the optimal dot point of $f = 0.55$; (b) scatter graph of the HR and SDHI with the optimal classification line.

the error rate of $4.3 \pm 3.7\%$ suggesting that the accuracy of the estimated HR is changing over time. This varying error rate phenomenon also can be seen in Fig. 6.

The evaluation result on the 125 data shows that the proposed method is able to measure the HR within 5 s with the accuracy of $96.2 \pm 2.5\%$ on average, and with the error of the HR estimation for the healthy subjects and dengue patients are of $3.3 \pm 2.1\%$ and $4.4 \pm 2.8\%$, respectively. This result is comparable to that of other state-of-the-art studies [18–20]. The comparison to the other methods is summarized in Table 3. Also, the result in our experiment in Section 3.1 suggests that the heartbeat signals from the dengue patients are less stable than those of the healthy subjects, leading less accurate in the HR estimation.

Using only the HR information, we performed classification on of the dengue patients and healthy subjects, and achieved the sensitivity of 82.4% and the specificity of 76%. On the other hand, we utilized LDA with two parameters: the HR and SDHI, and achieved the sensitivity of 86.3% and, the specificity of 86.9%. This result can be explained that, according to Bayes' theorem [39], the more independent information is provided, the higher sensitivity and specificity values we achieved.

Beside dengue fever, several infectious diseases, such as influenza and severe acute respiratory syndrome (SARS) can be identified via vital sign monitoring approach along with other clinical symptoms. In a study by Nishiura and Kamiya [3], the body temperature was used to classify infected subjects (H1N1), achieving a sensitivity of from 50.8 to 70.4% and, a specificity of from 63.6 to 81.7%. Sun et al. [40] utilized the HR and respiratory rate combined with the body temperature on classification of seasonal influenza in a period of 15 s, and obtained a sensitivity and a specificity of 85.6% and 88.6%, respectively. This result showed

that the proposed method using short duration of time achieved a similar accuracy. Yang et al. [36] used three parameters of the HR, respiratory rate and SDHI to classify dengue patients, using a machine learning-based methods. A classification accuracy of 98% was reported. Obviously, respiratory information enables a better accuracy in the classification; however, this method requires at least 30-s data length for the breathing circle analysis.

Recently, Covid-19 pandemic has been a global threat, which was first reported in Wuhan, China and has spread to more than 215 countries just within a few months. At least, ten million infected cases and five hundred thousands of deaths have been reported [41]. Several studies indicated that Covid-19 infected patient suffers a symptom of elevated heart rate [42–44]. Although the classification using contactless method such as CW Doppler radar is not perfect, fast screening to identify the infected patient for public health quarantine may have a large potential on preventing the rate of infection. Due to the simplicity and convenience of the CW Doppler radar monitoring technology, the device may be used in several public areas such as train station and airport.

Although EMD-CWT method has shown an adequate accuracy in estimating the HR, there are some weaknesses in this study. Firstly, in the experiment of this study, we utilized the selected stable segments of the received signal. When the subject continually moves and the breathing circle is not in harmony, the instability in the received signal may result in incorrect cardio-vascular pulses estimation. This is a challenge because the requirement for the subject staying still with breath holding seems to be infeasible. The further studies on stabilizing the signal and removing the body motion noises may deal with the same situation. Besides, the number of subjects participated in the study is limited; nevertheless, given the same experiment setup, we expect that the other studies will obtain the results which are similar to what we achieved. Recently, machine learning-based classification methods for infected diseases screening have shown promising results [36]. Though, building a large dataset to train the classification model involves a lot of data collection and manual annotations, which may be limited by administrative policy in different countries. Thus, we believe that researches based on traditional signal processing methods for the HR estimation still have a significant impact on the field of infected disease screening.

5. Conclusions

In conclusion, we proposed a method for the contactless cardio-vascular pulses estimation using the CW Doppler radar signals in a short time (5 s) with a reasonable accuracy (96.2%). The evaluation on dataset of both dengue patients and healthy subjects showed that the method outperforms the conventional method STFT. Moreover, we applied LDA with the HR and SDHI information to classify the dengue

Table 3

Comparison of the estimated HR accuracy of some state-of-the-art contactless systems based on CW Doppler radar.

Citation	System and method	Number of subjects	Analysis time	Accuracy
Yang et al. [19]	24 GHz CW Doppler radar, BPF + peak detection in time domain	7 (healthy)	A breathing cycle (approximately 5 s)	$R = 0.92$
Li and Lin [18]	5.8 GHz CW Doppler radar, time window variation + WT	3 (healthy)	Time variation 3–5 s	96.5%
Tu and Lin [20]	5.8 GHz CW Doppler radar, time window variation + FFT	4 (healthy)	Time variation 2–5 s	96.6%
This work	10 GHz CW Doppler radar, EMD-CWT	125 (dengue, healthy)	5 s	$R = 0.97$ $96.2 \pm 2.5\%$

patients. The results showed that using the HR and SDHI yields higher classification accuracy than that of using only the HR, achieving the sensitivity of 86.3% and the specificity of 86.9%. We believe that this promising result enables other further studies for the short time HR estimation using the contactless device.

Author contributions

Conceptualization: L.M.H., G.S., T.D.T., K.I. and N.L.T.; methodology: N.D.C., L.M.H. and G.S.; software: N.D.C. and L.Q.A.; validation: N.D.C. and L.Q.A.; formal analysis: N.D.C. and L.Q.A.; investigation: all of the authors; resources: N.D.C., L.M.H., G.S., L.Q.A. and N.V.T.; data curation: L.M.H. and G.S.; writing – original draft preparation: N.D.C., L.M.H., G.S., L.Q.A., P.V.H., T.A.V., T.T.H., T.D.T., N.V.T. and N.L.T.; visualization: L.Q.A.; supervision: K.I. and N.L.T.; project administration: L.M.H. and G.S.; funding acquisition: L.M.H. and G.S. All authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare no conflict of interest.

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